Rakshit Ratan Pathak

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SUMMARY

Final-year B.Tech student (CGPA 8.3) who builds and evaluates ML and LLM systems. Built the RAG layer of a healthcare LLM product at aiDataWorks. My ML projects report multi-seed results with released code and data, and back three IEEE Access manuscripts and five published patents.

EDUCATION

Vellore Institute of Technology, Vellore | B.Tech, Computer Science and Business Systems | **CGPA 8.3** 2023–2027

Class XII (CBSE), Delhi Public School Ranchi: 87% | Class X (CBSE), DAV Jharsuguda: 99.8%

Relevant coursework: Social, Text and Media Analytics (text mining, topic modelling), Data Structures and Algorithms, Design and Analysis of Algorithms, DBMS, Information Systems Audit and Control.

EXPERIENCE

Software Engineering Intern, aiDataWorks (remote) May–Jul 2026

- Built the generative AI layer of a healthcare product: designed and iterated prompts, retrieval and context for a RAG pipeline on OpenAI GPT APIs and enterprise Google Cloud, testing outputs to keep answers grounded.
- Wrote API- and database-level test cases for the LLM features and re-verified fixes with developers.
- Wrote data-governance and security policies for three client engagements, covering mainframe health-insurance data, cloud shop-floor data and a credential migration.

Computer Science Intern, THDC India Limited (Khurja Super Thermal Power Project) May–Jun 2025

- Worked on SQL and SAP ERP data workflows in Python and Pandas, plus SCADA/PLC systems, plant networking and cybersecurity controls.

PROJECTS

Opinion Lineage System: self-correcting sentiment (Python, scikit-learn) Paper code | RAG

- A calibrated gradient-boosted model infers which posts are reposts, paraphrases, coordinated copies or AI rewrites of earlier opinions on real tweets. It cut provenance error 2.5x vs the best heuristic pipeline (0.0030 vs 0.0075) at 40% contamination.
- A lineage-aware RAG layer (one evidence slot per lineage, in-place correction on retraction) cut answer error from 0.098 to 0.061 vs naive RAG.

agentic-bug-repair: autonomous bug-fixing agent (Java, hybrid retrieval) GitHub

- Localizes bugs with a code graph plus text retrieval, proposes diffs, verifies them in a sandbox and feeds failures back for up to 3 rounds. Fixed 11 of 17 seeded bugs with the offline synthesizer; the LLM path is built but not yet benchmarked.

RailConflict-PSCL: railway collision-risk assessment (Python, conformal prediction) GitHub

- Added two safety layers to a collision-risk classifier: per-class conformal escalation and a braking-physics shield. It cut missed-Critical encounters from 9.9% (gradient boosting) to 0.7% across 10 seeds, at a 9% false-emergency rate.

TrustDecay-TKG: IT-asset audit prioritization (Python, temporal knowledge graphs) GitHub

- Forecasts how far each IT asset's trust has decayed from audit and security events. It reached 54% precision at a 10% audit budget vs 16% for rotation audits, and ablations showed the gain came from temporal features.

VeriCloud: news claim verification (Python, scikit-learn) GitHub

- BM25 + TF-IDF retrieval, a stance model and learned source reliability, fused into calibrated verdicts that abstain on thin evidence. Under simulated sybil attacks it held macro-F1 at 0.79 vs 0.70 for the best baseline ($p = 0.008$).

Risk-Constrained Emergency Control (Python, scikit-learn) Live demo | GitHub

- Safety-gated gradient-boosting screener that blocks unsafe control actions, deployed as a web console on Vercel.

Also on GitHub: cuda-gpu-bench (CUDA kernels), minic (a C++17 compiler) and mini-kv-store (an LSM-tree store in Java).

SKILLS

- **Languages:** Python, SQL, Java, C++, TypeScript
- **ML and evaluation:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, gradient boosting, conformal prediction, calibration, knowledge graphs, significance testing, Power BI
- **GenAI:** LLMs, OpenAI GPT APIs, prompt engineering, RAG, grounding and guardrails, embeddings, hybrid retrieval (BM25, TF-IDF)
- **Engineering:** Flask, FastAPI, Streamlit, Node.js, PostgreSQL, Docker, Git, GitHub Actions, pytest, Google Cloud, Oracle Cloud (OCI), Vercel

PATENTS, RESEARCH AND LEADERSHIP

- **Six patent applications** through VIT's IPR Cell, five published: opinion-lineage sentiment correction (NLP), IS trust-decay prediction (temporal knowledge graphs), wildfire risk prediction (ML ensemble), mine tremor detection (edge AI and IoT), risk-constrained emergency control (safety-gated ML). Filed: railway collision avoidance.
- **IEEE Access manuscripts** on railway collision risk, IS trust decay and opinion lineage, each with open code and data and a one-command reproducible run.
- **Certifications:** OCI Generative AI Professional, OCI AI Foundations, Google Cloud Core Infrastructure, Google AI Essentials.
- Secretary, IEEE Power & Energy Society student chapter, VIT. Karate black belt and assistant instructor.